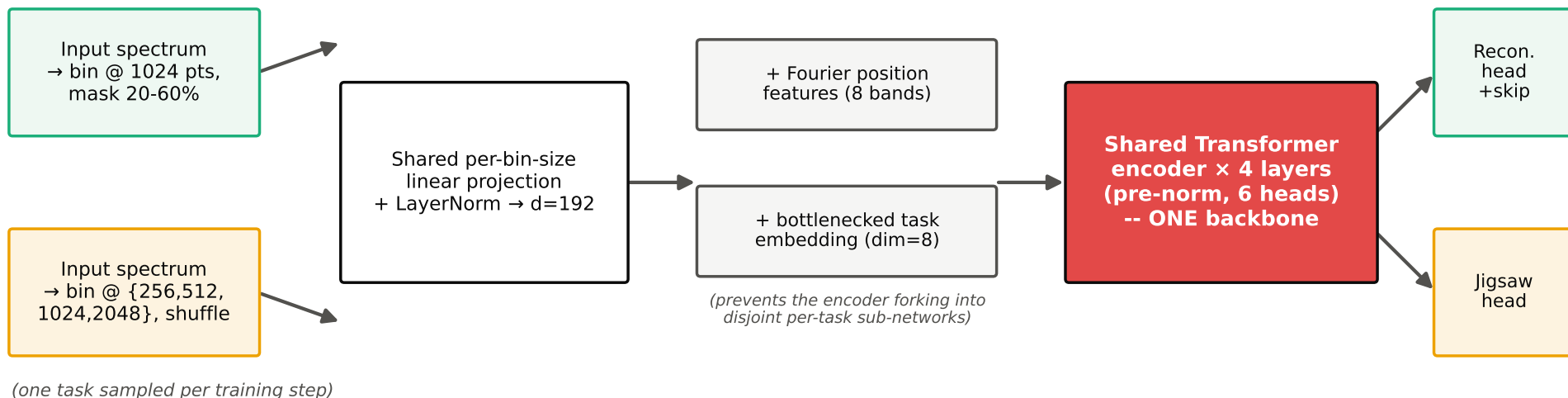


Joint masked + jigsaw SSL

`train_joint_ssl.JointNMRSSLModel -- shared_d_model=192, 4 layers, 6 heads, FFN=768, task_embed_dim=8 (bottlenecked), Fourier position bands=8`



Masked loss: MSE over only the top 17.5% highest-magnitude bins per spectrum (`peak_top_fraction`) -- avoids the loss being trivially satisfied by predicting a flat baseline. Jigsaw loss: cross-entropy, same as the standalone jigsaw model.

